Semi-Supervised Dynamic Ensemble Learning With Balancing Diversity and Consistency for Hyperspectral Image Classification

Hongliang Lu, Hongjun Su, *Senior Member, IEEE*, Pan Zheng, Yihan Gao, Hengyi Zheng, Zhaohui Xue, *Member, IEEE*, Weiwei Sun, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Hyperspectral coastal wetland classification requires an extensive quantity of labeled samples, which are hard to acquire. Therefore, a novel semi-supervised dynamic ensemble learning (SSDEL) framework is proposed to overcome the limitations of labeled samples in wetland hyperspectral classification. First, a collaborative relationship is established between labeled and unlabeled samples in the sample augmentation stage. Based on this relationship, unlabeled samples were assigned to the region to which the most similar samples belonged. Then, multiple classifiers are trained using labeled samples and predict unlabeled samples in the same region to obtain higher confidence pseudo-label results. Second, based on the assumption that different classifiers should produce similar classification results for a specific target sample, an objective function is designed to unify the classification behavior of multiple classifiers. The representation coefficients of multiple classifiers in the same region are constrained by optimizing the objective function through the l_2-norm. Finally, a complete SSDEL framework is constructed by applying consistency learning again to the augmented samples. The proposed method is evaluated using three wetland hyperspectral images of China, and the experimental results demonstrate its effectiveness.**

***Index Terms*— Classification, hyperspectral, limited samples, semi-supervised learning (SSL), wetlands.**

I. Introduction

COASTAL wetland land cover classification is related to the categorization of various landforms present in coastal regions, including beaches, water bodies, and other distinct types [1]. It is aimed at enhancing the understanding of wetland structure and functioning by systematically classifying different wetland types [2]. Therefore, the fine classification of coastal wetland land cover is of immense scientific significance and practical utility in wetland preservation and protection [3]. Hyperspectral data offer a wealth of information on ground objects due to its extensive collection of contiguous narrow spectral bands, rendering it a reliable technical approach for fine wetland monitoring [4], [5], [6], [7], [8], [9], [10], [11]. A large number of training samples are needed for fine classification of wetlands using hyperspectral data [12], [13]. Unfortunately, due to the extensive diversity of wetland species and their unique geographical distributions, obtaining a sufficient quantity of viable labeled samples presents a challenge [14], [12], [15], [16], [17], [18]. Therefore, addressing the issue of inadequate labeled samples is the key to wetland classification using hyperspectral image (HSI) [19].

Semi-supervised learning (SSL) [20] is one of the techniques to address the constraint of limited samples by leveraging a limited number of labeled samples alongside a vast number of unlabeled samples [21], [22]. Unlike supervised learning, SSL exploits information from unlabeled data to improve model performance amid label scarcity [23]. The fundamental assumption of SSL is that labeled and unlabeled samples have the same data distribution, which enriches the model's generalization ability.

The first category of SSL approaches is characterized by generative models. In generative models, the model seeks to understand the overall distribution of the data, encompassing both labeled and unlabeled samples. By learning this distribution, generative models can generate pseudo-labels for unlabeled samples, thereby expanding the training data and improving classification performance [24]. Distributional properties in unlabeled samples are captured to enhance generalization. Zhong et al. [25] introduced a novel approach background-anomaly separability with generative adversarial network (BASGAN), which uses a generative adversarial network (GAN) to improve the separability of background and anomaly in HSIs, achieving great results in real data experiments. However, accurate representation of the data distribution is hindered by challenges such as mode collapse and mode degeneracy, degrading the effectiveness of SSL. Qin et al. [26] presented DGAN, a distance constraint-based GAN method for HSI classification, addressing the challenges

Manuscript received 22 January 2024; revised 26 February 2024; accepted 30 March 2024. Date of publication 8 April 2024; date of current version 19 April 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42122008 and Grant 42371327, in part by the Postgraduate Research and Practice Innovation Program of Jiangsu Province under Grant KYCX22_0662, and in part by the Fundamental Research Funds for the Central Universities under Grant B230205051. *(Corresponding author: Hongjun Su.)*

Hongliang Lu, Pan Zheng, Yihan Gao, and Hengyi Zheng are with the School of Earth Sciences and Engineering, Hohai University, Nanjing 211100, China.

Hongjun Su and Zhaohui Xue are with the College of Geography and Remote Sensing, Hohai University, Nanjing 211100, China (e-mail: hjsu@hhu.edu.cn).

Weiwei Sun is with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo 315211, China.

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

Digital Object Identifier 10.1109/TGRS.2024.3386400

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

of limited labeled data and class imbalance. DGAN uses convolution autoencoders to extract features and leverages distance constraints to generate synthetic HSI samples, achieving great performance in accuracy on multiple datasets with minimal labeled data. However, accurate learning of the data distribution is hindered by challenges such as mode collapse and mode degeneracy, degrading the effectiveness of SSL.

The second category involves graph models, where graph structures are utilized to represent relationships and similarities in the data [27]. Label propagation algorithms and graph convolutional networks (GCNs) belong to this type [28]. Information propagation or graph convolution is employed for feature extraction, leveraging information in unlabeled samples to improve model performance. Su et al. [10] introduced a novel GNN approach for HSI classification. It utilizes autoregressive moving average filters instead of spectral filters, incorporates a dense structure for locality preservation, and employs layerwise context-aware learning, outperforming existing state-of-the-art methods on real HSI datasets. However, applying these models to large-scale data encounters computational complexity challenges, and performance heavily depends on the accuracy and connectivity of the graph structure and edges.

The third category of SSL is distribution-based methods, assuming that both labeled and unlabeled samples are derived from the same data distribution [29], [30]. Performance is enhanced by matching unlabeled sample distributions with labeled samples or incorporating unlabeled samples into auxiliary training. Zhang et al. [29] proposed an SSL HSI classification method based on modified co-training with spectral and spatial views, demonstrating its superiority, particularly when labeled samples are extremely limited. However, the effectiveness of these methods heavily relies on the fidelity of the assumption, and violations can significantly degrade their efficacy. Additionally, distributional assumption-based methods are sensitive to noise and outliers, leading to impaired performance if anomalous samples interfere with the training process [31], [32].

To solve the limitations of distribution-based approaches in SSL, this article proposes a novel semi-supervised dynamic ensemble learning (SSDEL) framework. It has three key components. First, in the sample augmentation stage, a collaborative relationship is established between labeled and unlabeled samples, and unlabeled samples were assigned to the region to which the most similar samples belonged. Then, multiple classifiers are trained using labeled samples and predict unlabeled samples in the same region to obtain higher confidence pseudo-label results. Second, the classification results of different classifiers can be regarded as expressions of different modalities for the target region. Meanwhile, for the same sample, the results predicted by different classifiers should be consistent. Based on this assumption, an objective function that constrains the behavior of multiple classifiers through an l_2-norm minimization is designed. Specifically, the relationship between unlabeled and labeled samples is leveraged to update the training samples and the representation coefficients in this function. Then, the multiclassifier fusion results are obtained through the consistency representation coefficient. Finally, a complete SSDEL framework is constructed by applying consistency learning again to the augmented samples. This framework combines methods of sample augmentation and consistent learning to effectively utilize both labeled and unlabeled samples for SSL, improving model performance and generalization. The main contributions can be summarized as follows.

1) *New Distance Metric:* The characteristics of wetland hyperspectral data include a wide variety of land cover types, extensive distribution, and a relatively scattered nature. Therefore, a single spatial distance is unable to effectively capture the complex relationships among wetland samples. To address this issue, a collaborative relationship is established between labeled and unlabeled samples to obtain pseudo-label results with higher confidence. Then, the information of unlabeled samples can be effectively utilized, and the accuracy of SSL can be improved.
2) *Classifier Consistency Method:* In contrast to other ensemble learning methods that use static classifier integration techniques like voting or weighting, an objective function for a dynamic classifier ensemble is proposed in this article. For this objective function, the behavior of multiple classifiers using the l_2-norm is proposed in order to achieve the consistency of multiclassifier results in ensemble learning. The consistency among base classifiers is improved and the diversity is reduced by this method, addressing the potential inconsistency issues in distribution-based methods.
3) *Adaptive Semi-Supervised Ensemble Learning Framework:* Unlike other classification methods, the theory of dynamic ensemble selection (DES) is introduced into SSL in this article. The diversity and consistency in ensemble learning are considered simultaneously based on the proposed distance metric and consistency constraint method. The SSDEL framework effectively balances the requirements of sample expansion, classifier diversity, and consistency, considering multiple key factors in SSL.

The remainder of this article is organized as follows. In Section II, the related works are introduced. In Section III, the proposed distance metric and SSDEL are presented. The results of the experiment are demonstrated in Section IV. In Section V, the parameter sensitivity, generalizability, effectiveness, ablation analysis, and advancement of the proposed algorithms are discussed in detail. Finally, the conclusions of this study are drawn in Section VI.

II. Related Work

A. Base Classifiers Used in This Study

Collaborative representation-based classification (CRC) [33], [34], [35] was initially introduced for face recognition and has subsequently gained popularity in HSI classification due to its remarkable performance. Consider a training sample denoted by $\boldsymbol{X} = [x_1, \ldots, x_n]$, which consists of n features. The training set is a matrix $\boldsymbol{X}$ of size $n \times N$, where N is the number of training samples. The testing samples, denoted

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

as $\boldsymbol{Y} = [y_1, \ldots, y_n]$, are assigned with class labels from the set $\{1, \ldots, c\}$, where c represents the total number of classes. The testing set is a matrix $\boldsymbol{Y}$ of size $n \times M$, where M is the number of testing samples. The collaborative representation (CR) coefficient $\boldsymbol{\alpha}$ is obtained

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}^*} ||\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\alpha}^*\|_2^2 + \lambda \|\boldsymbol{\alpha}^*\|_2^2 \tag{1}$$

where $\|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\alpha}^*\|_2^2$ is the reconstruction error, λ is the regularization parameter, and $\lambda\|\boldsymbol{\alpha}^*\|_2^2$ controls the complexity of $\boldsymbol{\alpha}^*$. By zeroing out the partial derivative, the analytical solution can be derived

$$\boldsymbol{\alpha} = \left(\boldsymbol{X}^{\mathrm{T}}\boldsymbol{X} + \lambda \boldsymbol{I}\right)^{-1}\boldsymbol{X}^{\mathrm{T}}\boldsymbol{Y} \tag{2}$$

where $\boldsymbol{I}$ denotes an identity matrix.

In order to address the issue of linear inseparability of pixels in image classification, one approach, i.e., kernel collaborative classification (KCRC) [36], [37], is proposed. The optimization problem can be reformulated as

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}^*} ||\boldsymbol{\Phi}(\boldsymbol{Y}) - \boldsymbol{\Phi}(\boldsymbol{X})\boldsymbol{\alpha}^*\|_2^2 + \lambda \|\boldsymbol{\alpha}^*\|_2^2. \tag{3}$$

Similar to (2), the analytical solution can be rewritten as

$$\boldsymbol{\alpha} = (\boldsymbol{\Phi}\ (\boldsymbol{X})^T \boldsymbol{\Phi}(\boldsymbol{X}) + \lambda \boldsymbol{I})^{-1}\boldsymbol{\Phi}(\boldsymbol{X})^T \boldsymbol{\Phi}(\boldsymbol{Y}) \tag{4}$$

where $\Phi(\boldsymbol{Y})$ and $\Phi(\boldsymbol{X})$ represent the testing samples. According to the kernel trick, their inner product equals

$$\kappa(x_i, x_i) = \boldsymbol{\Phi}(x_i)^{\mathrm{T}}\Phi(x_i) \tag{5}$$

where $\kappa(.,.)$ is a kernel function, e.g., a Gaussian function. Similar to (4), the analytical solution can be rewritten as

$$\boldsymbol{\alpha} = (\kappa(\boldsymbol{X}, \boldsymbol{X}) + \lambda \boldsymbol{I})^{-1}\kappa(\boldsymbol{X}, \boldsymbol{Y}) \tag{6}$$

where $\kappa(\boldsymbol{X}, \boldsymbol{X})$ is an $N \times N$ Gram matrix and $\kappa(\boldsymbol{X}, \boldsymbol{Y})$ is equivalent to $\boldsymbol{\Phi}(x_i)^{\mathrm{T}}\boldsymbol{\Phi}(y_j)$ based on the kernel trick. The information used for classification usually is the residual corresponding to the c th class

$$r_k(\boldsymbol{y}) = \|\boldsymbol{y} - \boldsymbol{X}_{\boldsymbol{c}}\boldsymbol{\alpha}_c\|_2^2 \tag{7}$$

$$\text{class}(\boldsymbol{y}) = \arg\min_c r_c(\boldsymbol{y}) \tag{8}$$

where $r_c(y)$ is the difference between the testing sample and its reconstructed value. Three heterogeneous CR-based classifiers, namely, CRC, KCRC, and probabilistic CR (ProCRC) [38], [39], are selected as the base classifiers for semi-supervised dynamic ensemble in this article.

III. Proposed Method

In this section, the structure of the proposed SSDEL is introduced. The novel weighted co-distance (WCD) method is then presented, and its application in attaining high-confidence and high-information sample augmentation within an active learning framework is elucidated. Following this, the consistent coefficient optimization (CCO) algorithm is explained in detail. Finally, an analysis of the SSDEL strategy is conducted. The algorithm's detailed description and derivation are as follows.

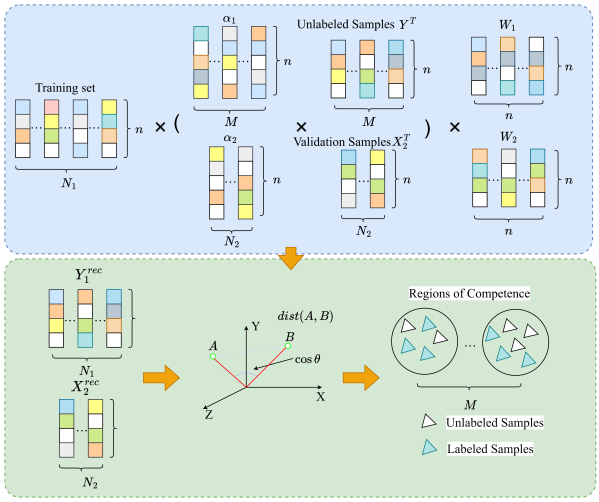

Fig. 1. Flowchart for the proposed WCD. ($\boldsymbol{\alpha_1}$ and $\boldsymbol{\alpha_2}$ are derived from (9) and (10); $\boldsymbol{W_1}$ and $\boldsymbol{W_2}$ are derived from (15) and (16); and the training and validation sets consist of five samples per class each, with the remaining samples reserved for testing).

A. Weighted Co-Distance

1) Motivation of Introducing a Novel Distance Metric: DES [40], [41], [42] is a method employed in ensemble learning to enhance classification performance by the selection of a subset of base classifiers that are deemed most suitable for the current samples. In DES, the establishment of a region of competence (RoC) entails the utilization of measurement methods, typically distance-based or similarity-based, to establish connections between unknown samples and training samples, leading to the partitioning of the sample space into separate regions. By selecting well-performing base classifiers within each region, the classification performance and generalization capacity of the ensemble model are effectively improved. However, traditional RoC partitioning is often reliant on the k-nearest neighbor algorithm, which only considers the spatial distance between samples [43], [44]. Nevertheless, relying on spatial distance is inadequate for effectively capturing the intricate relationships among samples. Therefore, a novel distance metric WCD is proposed in this study, as illustrated in Fig. 1. The main idea of WCD is to simultaneously establish relationships among training, validation, and testing samples, providing a comprehensive and reliable definition of ability regions. Additionally, CR is employed to accurately describe the collaborative relationships between samples, leading to a more appropriate understanding of the dynamic ensemble.

2) Definition of WCD: The specific process of WCD is illustrated in Fig. 1 and can be divided into two main parts. First, it involves reconstructing test and validation samples using CR coefficients and feature weights, establishing their relationships with training samples. Then, the distance between the reconstruction results is measured using cosine distance. The detailed definition process is as follows.

Let $\boldsymbol{X}_1 = [x_{1,1}, \ldots, x_{1,n}]$ be a training sample characterized by n features. The training set is represented by $\boldsymbol{X_1}$, which is an $n \times N_1$ matrix and N_1 is the number of training samples.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

Similarly, $\boldsymbol{X}_2 = [x_{2,1}, \ldots, x_{2,n}]$ represents a validation sample. The validation set is denoted by $\boldsymbol{X_2}$, which is an $n \times N_2$ matrix and N_2 is the number of validation samples. The testing samples are defined as $\boldsymbol{Y} = [y_1, \ldots, y_M]$, where $\boldsymbol{y}_i \in \{1, \ldots, c\}$, and c represents the number of classes. The representation coefficients for the testing samples $\boldsymbol{\alpha}_1$ and validation samples $\boldsymbol{\alpha}_2$ can be obtained as

$$\boldsymbol{\alpha}_1 = \arg\min_{\boldsymbol{\alpha}^{1*}} ||\boldsymbol{Y} - \boldsymbol{X}_1{\boldsymbol{\alpha}_1}^*\|_F^2 + \lambda \|{\boldsymbol{\alpha}_1}^*\|_2^2 \tag{9}$$

$$\boldsymbol{\alpha}_2 = \arg\min_{\boldsymbol{\alpha}^{2*}} ||\boldsymbol{X}_2 - \boldsymbol{X}_1{\boldsymbol{\alpha}_2}^*\|_F^2 + \lambda \|{\boldsymbol{\alpha}_2}^*\|_2^2 \tag{10}$$

where $\|\cdot\|_2$ represents the l_2-norm and λ is the regularization parameter. It is assumed that the distribution of the validation and testing samples can be adequately represented by $\boldsymbol{\alpha_1}$ and $\boldsymbol{\alpha_2}$.

In practical applications, the introduction of feature weighting allows for a more accurate assessment of the significance associated with each feature, thereby facilitating a comprehensive understanding of the data [45]. Thus, in this article, feature weights are obtained through reconstruction error, and new weighted representation coefficients are obtained accordingly. The reconstruction error of testing samples $\boldsymbol{Y}$ and validation samples $\boldsymbol{X_2}$ can be defined as

$$\boldsymbol{E_1} = \|\boldsymbol{Y} - \boldsymbol{Y^{rec}}\|_2 = \|\boldsymbol{Y} - \boldsymbol{Y^*W}^{\mathrm{T}}\|_2 = \left\|\boldsymbol{Y} - \boldsymbol{X_1}(\boldsymbol{\alpha_1 Y}^{\mathrm{T}})\boldsymbol{W_1}^{\mathrm{T}}\right\| \tag{11}$$

$$\begin{aligned}\boldsymbol{E_2} &= \|\boldsymbol{X_2} - \boldsymbol{X_2^{rec}}\|_2 = \|\boldsymbol{X_2} - \boldsymbol{X_2^*W}^{\mathrm{T}}\|_2 \\ &= \left\|\boldsymbol{X_2} - \boldsymbol{X_1}(\boldsymbol{\alpha_2 X_2}^{\mathrm{T}})\boldsymbol{W_2}^{\mathrm{T}}\right\|\end{aligned} \tag{12}$$

where $\boldsymbol{E_1}$ and $\boldsymbol{E_2}$ represent the reconstruction error of testing samples $\boldsymbol{Y}$ and validation samples $\boldsymbol{X_2}$, and $\boldsymbol{W}$ is weighted representation coefficients of size $n \times n$. Notably, the reconstruction of $\boldsymbol{Y}$ and $\boldsymbol{X_2}$ is not directly utilized by $\boldsymbol{X_1\alpha_1}$ and $\boldsymbol{X_1\alpha_2}$. This is because differences may exist between the training and testing samples, and the direct utilization of $\boldsymbol{X_1\alpha_1}$ as $\boldsymbol{Y^{rec}}$ may result in larger reconstruction errors, as the features of the testing samples may not be fully captured by $\boldsymbol{X_1\alpha_1}$. Moreover, since $\boldsymbol{\alpha_1}$ is calculated within the training sample space, some specific information regarding the testing samples may be lost by $\boldsymbol{X_1\alpha_1}$. Thus, $\boldsymbol{X_1}(\boldsymbol{\alpha_1 Y}^{\mathrm{T}})\boldsymbol{W_1}$ is employed as $\boldsymbol{Y^{rec}}$ in this part, where $\boldsymbol{\alpha_1}$ denotes the CR coefficient and $\boldsymbol{W_1}$ denotes the feature weight. By projecting the testing samples $\boldsymbol{Y}$ onto the training sample space using the CR coefficient $\boldsymbol{\alpha_1}$, the matrix $\boldsymbol{\alpha_1 Y}^{\mathrm{T}}$ is obtained, and then, it is weighted by utilizing the feature weight $\boldsymbol{W_1}$, so as to obtain the reconstructed testing sample $\boldsymbol{Y^{rec}}$. Based on this conclusion, the same approach is applied to validation samples $\boldsymbol{X_2}$.

To solve for the feature weight $\boldsymbol{W}$, it is required to minimize the reconstruction error. Thus, the following optimal solution problem can be formed as:

$$\arg\min_{\boldsymbol{W_1}} \left\|\boldsymbol{Y} - \boldsymbol{X_1}(\boldsymbol{\alpha_1 Y}^{\mathrm{T}})\boldsymbol{W_1}^{\mathrm{T}}\right\|_2^2 \tag{13}$$

$$\arg\min_{\boldsymbol{W_1}} \left\|\boldsymbol{X_2} - \boldsymbol{X_1}(\boldsymbol{\alpha_2 X_2}^{\mathrm{T}})\boldsymbol{W_2}^{\mathrm{T}}\right\|_2^2. \tag{14}$$

The solution is expressed as follows:

$$\boldsymbol{W_1} = \left\{\left[\boldsymbol{X_1}(\boldsymbol{\alpha_1 Y}^{\mathrm{T}})\right]^{\mathrm{T}}\boldsymbol{X_1}(\boldsymbol{\alpha_1 Y}^{\mathrm{T}})\right\}^{-1}\left[\boldsymbol{X_1}(\boldsymbol{\alpha_1 Y}^{\mathrm{T}})\right]^{\mathrm{T}}\boldsymbol{Y} \tag{15}$$

$$\boldsymbol{W_2} = \left\{\left[\boldsymbol{X_1}(\boldsymbol{\alpha_2 X_2^T})\right]^{T}\boldsymbol{X_1}(\boldsymbol{\alpha_2 X_2^T})\right\}^{-1}\left[\boldsymbol{X_1}(\boldsymbol{\alpha_2 X_2^T})\right]^{T}\boldsymbol{X_2}. \tag{16}$$

Then, the reconstructed testing samples $\boldsymbol{Y^{rec}}$ and validation samples $\boldsymbol{X_2^{rec}}$ after feature weighting can be obtained. The cosine similarity is proposed in this study to measure the similarity between $\boldsymbol{Y^{rec}}$ and $\boldsymbol{X_2^{rec}}$, thus establishing a connection between the validation and testing samples. The characteristics and advantages of cosine similarity are that it is unaffected by the length of the vectors, easy to calculate, interpretable, etc. The cosine distance calculation between $\boldsymbol{Y^{rec}}$ and $\boldsymbol{X_2^{rec}}$ is as follows.

Let $\boldsymbol{y}^{(i)}$ be the ith row of $\boldsymbol{Y^{rec}}$, and $\boldsymbol{x}_2^{(j)}$ be the jth column of $\boldsymbol{X_2^{rec}}$. The weighted cosine similarity (WCS) between $\boldsymbol{y}^{(i)}$ and $\boldsymbol{x}_2^{(j)}$ is

$$\mathrm{WCS}\left(\boldsymbol{y}^{(i)}, \boldsymbol{x}_2^{(j)}\right) = \frac{\boldsymbol{w}_1^{(i)}\boldsymbol{y}^{(i)} \cdot \boldsymbol{w}_2^{(i)}\boldsymbol{x}_2^{(j)}}{\left(\left\|\boldsymbol{y}^{(i)}\right\| \cdot \left\|\boldsymbol{x}_2^{(j)}\right\|\right)} \tag{17}$$

where $\|.\|$ denotes the l_2-norm. Therefore, the WCS between each row of $\boldsymbol{Y^{rec}}$ and $\boldsymbol{X_2^{rec}}$ can be expressed as

$$\mathrm{WCS}\left(\boldsymbol{Y^{rec}}, \boldsymbol{X_2^{rec}}\right) = \frac{\boldsymbol{W_1 Y^{rec}} \cdot \boldsymbol{W_2 X_2^{rec}}}{\left(\|\boldsymbol{Y^{rec}}\| \cdot \left\|\boldsymbol{X_2^{rec}}\right\|\right)}. \tag{18}$$

Based on (18), a similarity matrix S of size $(M, N2)$ can be obtained

$$\boldsymbol{S} = \begin{bmatrix} d_{1,1} & \cdots & d_{1,N2} \\ \vdots & \ddots & \vdots \\ d_{M,1} & \cdots & d_{M,N2} \end{bmatrix} \tag{19}$$

where each row represents an index of a validation sample, and each column represents an index of a testing sample. By removing the maximum value of each row, we can obtain the similarity between each testing sample and its most similar validation sample, i.e., the corresponding ability region. Finally, $\boldsymbol{WCD}$ can be calculated from the similarity matrix $\boldsymbol{S}$ as

$$\boldsymbol{WCD} = \mathrm{argmax}\ \boldsymbol{S}. \tag{20}$$

Based on Fig. 1, upon sorting, the testing samples can be partitioned into the RoC of their most similar validation samples.

B. Consistent Coefficient Optimization (CCO)

1) Motivation of CCO: Consistency among classifiers is deemed crucial in the improvement of model performance and robustness [46]. In ensemble learning, the predictions of different classifiers can be regarded as diverse representations of the target samples. These classifiers may possess varying forms of representation and features, but it is desirable for their predictions of the same sample to be as consistent as possible. For instance, in image classification tasks, similar results should be produced from multiple classifiers for a given image. The primary challenge addressed by multiclassifier

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

consistency learning is how to unify these diverse representations. The motivation behind the proposed method is to address the challenges posed by classifier disparities and leverage the relationship between training samples $\boldsymbol{X_1}$ and testing samples $\boldsymbol{Y}$ to update training samples $\boldsymbol{X_1}$ and the representation coefficients α_i of the CR classifier.

2) Algorithm Details: The proposed optimization framework involves the minimization of the following composite objective function:

$$\arg\min_{\boldsymbol{X_1'}} \left\|\boldsymbol{Y}-\boldsymbol{X_1'}\boldsymbol{\alpha_i}\right\|_F^2 + \lambda_1\left\|\boldsymbol{\alpha_i}-\bar{\alpha}_i\right\|_2^2 + \lambda_2\left\|\boldsymbol{X_1'}-\boldsymbol{X_1}\right\|_2^2 \quad (21)$$

where $\boldsymbol{X_1'}$ is the updated training sample, $\boldsymbol{\alpha_i}$ is a matrix comprising the coefficients from M different classifiers, and the $\bar{\alpha}_i$ is the mean of $\boldsymbol{\alpha_i}$. The objective is to optimize the prediction performance by achieving consistency among classifiers. The alternating least-squares algorithm is employed, wherein the optimization of $\boldsymbol{X_1}$ and $\boldsymbol{\alpha_i}$ is performed alternately to attain consistency in $\boldsymbol{\alpha_i}$ and update $\boldsymbol{X_1}$. By incorporating a regularization term $\|\boldsymbol{\alpha_i}-\bar{\alpha}_i\|_2^2$, the method aims to encourage the consistency of $\boldsymbol{\alpha_i}$ values and exploit the relationship between $\boldsymbol{X_1}$ and $\boldsymbol{Y}$ when updating $\boldsymbol{X_1}$. The detailed derivation is provided as follows.

Step 1: Optimization of $\boldsymbol{X}$. With $\boldsymbol{\alpha_i}$ fixed, the objective function is formulated as a quadratic form in $\boldsymbol{X_1'}$, and the optimal solution is obtained through differentiation. The objective function is expanded as

$$L(\boldsymbol{X}) = \left\|\boldsymbol{Y}-\boldsymbol{X_1'}\boldsymbol{\alpha_i}\right\|_F^2 + \lambda_1\left\|\left(\boldsymbol{\alpha_i}-\bar{\alpha}_i\right)\right\|_2^2 + \lambda_2\left\|\boldsymbol{X_1'}-\boldsymbol{X_1}\right\|_2^2. \quad (22)$$

The following steps are as below.

1) While $\boldsymbol{\alpha_i}$ is held constant, the objective function is expanded as

$$(\boldsymbol{X}) = \mathrm{tr}\left[\left(\boldsymbol{Y}-\boldsymbol{X_1}\bar{\alpha}_i\right)^{\mathrm{T}}\left(\boldsymbol{Y}-\boldsymbol{X_1}\bar{\alpha}_i\right)\right] + \lambda_2\mathrm{tr}\left[\left(\boldsymbol{X_1'}-\boldsymbol{X_1}\right)^{\mathrm{T}}\left(\boldsymbol{X_1'}-\boldsymbol{X_1}\right)\right]. \quad (23)$$

2) The derivative of $L(\boldsymbol{X})$ with respect to $\boldsymbol{X'}$ is computed, and it is equated to zero

$$\partial L(\boldsymbol{X})/\partial \boldsymbol{X_1'} = -\bar{\alpha}_i^{\mathrm{T}}\left(\boldsymbol{Y}-\boldsymbol{X_1}\bar{\alpha}_i\right) + \lambda_2\left(\boldsymbol{X_1'}-\boldsymbol{X_1}\right) = 0. \quad (24)$$

3) The resulting equation is solved to obtain the optimal solution for $\boldsymbol{X_1'}$

$$\boldsymbol{X_1'} = \left(\boldsymbol{Y}\bar{\alpha}_i^{\mathrm{T}} + \lambda_2\boldsymbol{X_1}\right)\left(\bar{\alpha}_i\bar{\alpha}_i + \lambda_2\boldsymbol{I}\right)^{-1}. \quad (25)$$

Step 2: Optimization of $\boldsymbol{\alpha_i}$. With $\boldsymbol{X_1'}$ fixed, the objective function is expressed as a quadratic form in $\boldsymbol{\alpha_i}$, and its optimal solution is derived via differentiation. The objective function is given by

$$L(\boldsymbol{\alpha_i}) = ||\boldsymbol{Y}-\boldsymbol{X_1'}\boldsymbol{\alpha_i}||_2^2 + \lambda_1||\boldsymbol{\alpha_i} .- \bar{\alpha}_i||_2^2. \quad (26)$$

The subsequent steps are as follows.

1) Fixing $\boldsymbol{X_1'}$, the objective function is expanded as

$$L(\boldsymbol{\alpha_i}) = \mathrm{tr}\left[\left(\boldsymbol{Y}-\boldsymbol{X_1'}\boldsymbol{\alpha_i}\right)\left(\boldsymbol{Y}-\boldsymbol{X_1'}\boldsymbol{\alpha_i}\right)\right] + \lambda_1\mathrm{tr}\left[\left(\boldsymbol{\alpha_i}-\bar{\boldsymbol{\alpha}}_i\right)^{\mathrm{T}}\left(\boldsymbol{\alpha_i}-\bar{\boldsymbol{\alpha}}_i\right)\right]. \quad (27)$$

2) Compute the derivative of $L(\alpha_i)$ with respect to α_i and set it to zero

$$\partial L(\boldsymbol{\alpha_i})/\partial\boldsymbol{\alpha_i} = -\boldsymbol{X_1'}^{\mathrm{T}}\left(\boldsymbol{Y}-\boldsymbol{X_1'}\boldsymbol{\alpha_i}\right) + \lambda_1\left(\boldsymbol{\alpha_i}-\bar{\alpha}_i\right) = 0. \quad (28)$$

3) Solve the resulting equation to obtain the optimal solution for $\boldsymbol{\alpha_i}$

$$\boldsymbol{\alpha_i} = \left(\boldsymbol{X_1'}^{\mathrm{T}}\boldsymbol{X_1} + \lambda_1\boldsymbol{I}\right)^{-1}\left(\boldsymbol{X_1'}^{\mathrm{T}}\boldsymbol{Y} + \lambda_1\bar{\alpha}_i\right). \quad (29)$$

Step 3: Iterate between steps 1 and 2. $\boldsymbol{X_1'}$ and $\boldsymbol{\alpha_i}$ are alternately updated until a convergence criterion is met. This criterion may involve evaluating the change in the objective function until reaching a maximum number of iterations $\boldsymbol{T}$.

C. SSDEL Framework

1) Samples Select Based on WCD: In the first part, a diversified semi-supervised ensemble process was executed by performing pairwise combinations of three sets of base classifiers based on the WCD method (Fig. 2). The specific procedure is as follows.

1) *RoC Partitioning:* The relationship between testing and validation samples, as well as training samples, was established using the proposed WCD method. This facilitated the assignment of testing samples and validation samples to the same region. Within this region, testing and validation samples exhibited comparable classification performance.
2) *Base Classifier Evaluation:* The performance of each base classifier within the RoC region was assessed. In other wordy, predictions were made for the validation samples within the RoC region, considering the accuracy and confidence of the classifier's predictions. If a particular classifier performs well on the current sample, it is considered to possess commendable classification performance for unknown samples (testing samples) within that region.
3) *Base Classifier Subset Selection:* Based on the RoC evaluation results, a subset of well-performing base classifiers was chosen within the RoC region.
4) *Ensemble Model Prediction:* The selected subset of base classifiers was employed to predict new samples.
5) *Sample Augmentation:* Unknown samples of the same category were subjected to classification using pairwise classifiers to generate pseudo-labels. This signifies that unknown samples were augmented by incorporating pseudo-labeled samples, thus endowing them with category labels. The resulting pseudo-labeled samples were merged with the original training data to construct an augmented training set.

2) Triclassifier Ensemble Learning Based on CCO Method: In the second part, the ensemble was performed based on the augmented samples described above, with the aim of obtaining consistent classification results.

1) *Training Based on CR Classifiers:* Three groups of augmented training samples were used to train three base classifiers. This step is intended to learn different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

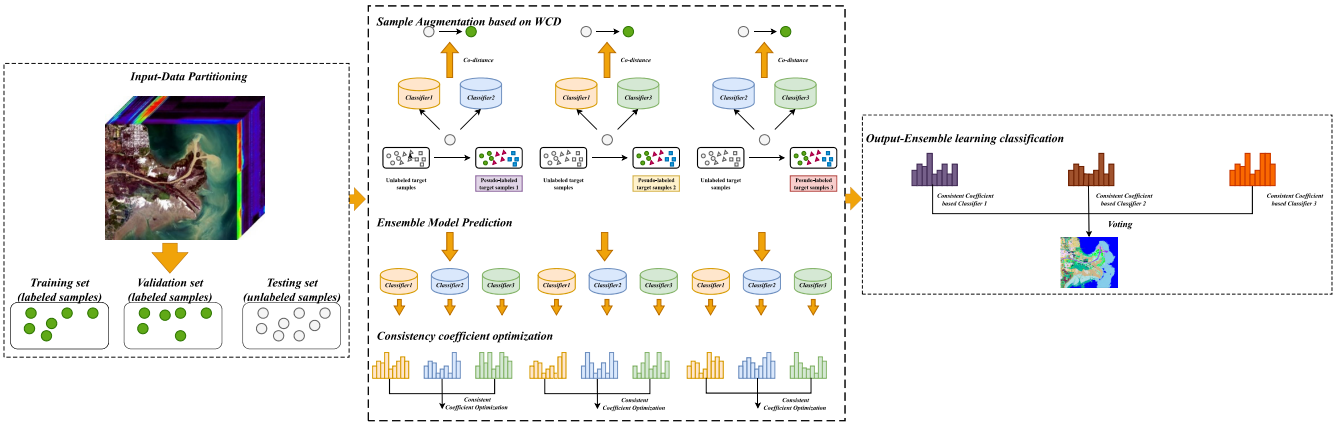

Fig. 2. Flowchart of the proposed SSDEL framework. (Part I: input-data partitioning: five samples per class were randomly selected from the hyperspectral data as training and validation samples, with the remaining portion designated as the testing samples; Part II: samples augmentation based on WCD: sample augmentation is conducted based on the proposed WCD; Part III: ensemble model prediction: the augmented samples obtained from Part II are utilized to train an ensemble of three classifiers; Part IV: CCO: consistency optimization is applied to the coefficients obtained from the training in Part III; and Part V: output-ensemble learning classification: in the final input section, residues are obtained and classified using coefficients optimized through consistency.)

representations of the data and obtain three sets of representation coefficients.

2) *Consistency Coefficient Optimization:* The proposed CCO method was employed to update the training samples and representation coefficients based on the original training samples and the newly obtained coefficients. This step was to ensure consistency in the classification results of different classifiers, thereby improving the generalization ability of the ensemble learning model.
3) *Ensemble Learning Classification:* Using the consistency-optimized representation coefficients, each group of training samples was classified to obtain three sets of classification results. Each set of results reflected the understanding of the data and classification decisions by different classifiers.
4) *Final Classification Result by Majority Voting:* The three sets of classification results were integrated using the majority voting method. The final classification label for each sample was determined through a majority vote.

IV. Experiments

This article utilizes three sets of Chinese coastland wetland hyperspectral data from the GF-5 satellite for classification experiments. The GF-5 satellite was launched on May 9, 2018, at Taiyuan Satellite Launch Center, Shanxi, China. It is the only hyperspectral satellite among the seven civilian satellites of the high-resolution Earth observation system major project. It is designed as a Sun-synchronous orbit with an orbital altitude of about 705 km.

A. Wetland Hyperspectral Datasets

The Yangtze River Delta of Jiangsu Province was used as the first wetland data to verify the effectiveness of the proposed algorithms. This data contains 268 spectral bands (after removing 62 bad bands) with wavelengths ranging from 0.39 to 2.5 μm. Its spatial size is 1160 $\times$ 1209. The details of nine classes in this HSI are described in Table I, and the images are shown in Fig. 3.

TABLE I
Elaborate on the Description of the Samples in Yangtze River Delta

Class	Ground object	Number
1	river	1273
2	mud flat (muf)	476
3	bare land (bal)	200
4	impervious surface (ims)	1046
5	tree	575
6	flood plain (flp)	435
7	farmland (far)	189
8	plant	455
9	lake	137
Total		4786

TABLE II
Elaborate on the Description of the Samples in Dafeng Natural Reserve

Class	Ground object	Number
1	sea	1020
2	river	276
3	building (bui)	89
4	aquaculture (acq)	746
5	farmland (far)	918
6	mud flat (muf)	528
7	tree	724
8	road	150
9	lake	79
Total		4530

The second wetland dataset chosen for the experiment is the Dafeng Natural Reserve. Comprising 256 spectral bands (with 74 bad bands removed), the wavelengths span from 0.39 to 2.5 μm. This dataset consists of 986 $\times$ 632 pixels and boasts a spatial resolution of 30 m. Class descriptions can be found in Table II, and images are visualized in Fig. 4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

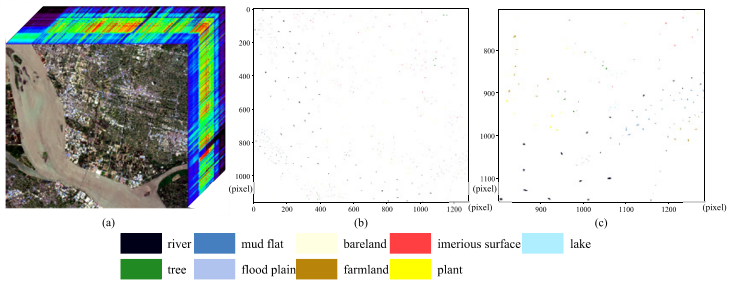

Fig. 3. (a) False-color image. (b) Corresponding ground truth. (c) Magnified section of the ground truth from Yangtze River.

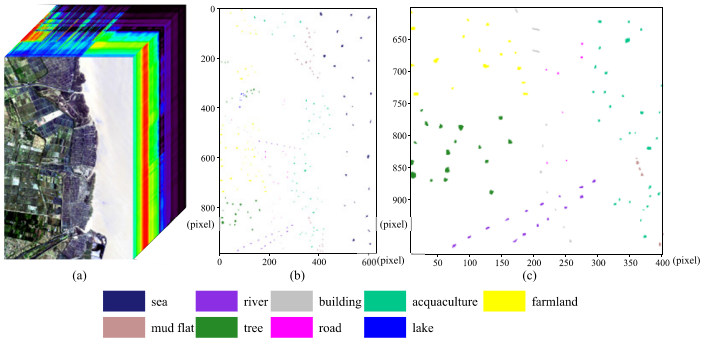

Fig. 4. (a) False-color image. (b) Corresponding ground truth. (c) Magnified section of the ground truth from Dafeng Natural Reserve dataset.

TABLE III
ELABORATE ON THE DESCRIPTION OF THE SAMPLES IN YELLOW RIVER DELTA

Class	Ground object	Number
1	acquaculture (acq)	393
2	rice	190
3	broomcorn (bro)	95
4	shallow sea(sha)	936
5	suaeda salsa (sus)	361
6	aquatic vegetation (aqv)	39
7	seep sea (ses)	796
8	building (bui)	83
9	locust	211
10	mud flat (muf)	553
11	reed	240
12	soybean (soy)	110
13	maize	96
14	spartina (spa)	200
15	river	469
16	salt marsh (sam)	595
17	intertidal saltwater (ins)	454
18	flood plain (flp)	68
19	tamarix (tam)	133
20	freshwater herbaceous marsh (fhm)	72
21	pond	377
Total		6491

The third dataset, comprising 1185 × 1324 pixels, covers the Yellow River Delta region. It includes spectral bands ranging from 0.4 to 2.5 μm, with 285 effective bands retained after eliminating 45 problematic ones. The dataset encompasses 21 distinct classes, and the spatial resolution is set at 30 m. Table III provides comprehensive information about each class, and you can visualize images of this dataset in Fig. 5.

B. Experimental Setup

The specific parameters of the proposed algorithms are set as follows. To verify the performance of the proposed algorithm in the case of small sample sizes, the training and validation data for all datasets in this article were randomly selected, with five samples per class being chosen. The regularization parameter for CCO $\lambda_1, \lambda_2 = \{0.1, 0.2, 0.3, 0.4, 0.5\}$.

C. Baseline and Comparative Algorithms

To mitigate the constraints of evaluating the proposed approach against a single classifier, enhancements were introduced to existing classifiers, i.e., the support vector machine (SVM) and random forest (RF) classifiers, through the integration of label propagation (LP) as a technique for augmenting samples. The LP-SVM and LP-RF classifiers were derived from this, offering a comparative analysis against methods based on LP. Furthermore, to evaluate the accuracy of proposed algorithms against state-of-the-art ensemble methods, the tritraining method was combined with classifiers such as Catboost [47], LightGBM [48], and GBDT [49]. In addition, the tritraining algorithm was integrated with the CR-based classifiers utilized in our proposed algorithm, showcasing the superior performance of our method in the context of SSL. Finally, some of the most advanced deep learning algorithms are also used as comparison algorithms, namely, GCN [50], graph attention network (GAT) [51], and graph sample and aggregation (GraphSAGE) [52].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

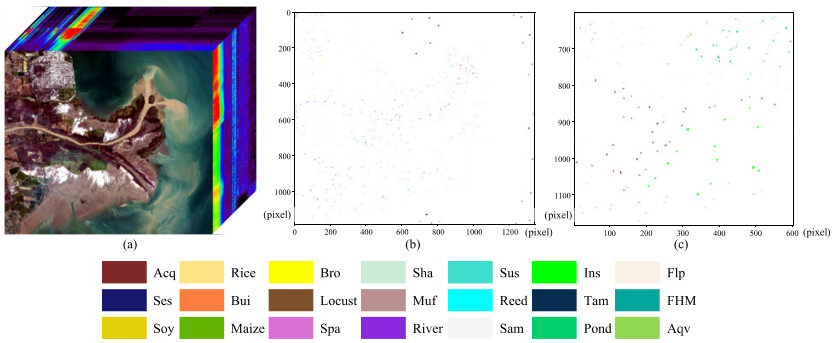

Fig. 5. (a) False-color image. (b) Corresponding ground truth. (c) Magnified section of the ground truth from Yellow River.

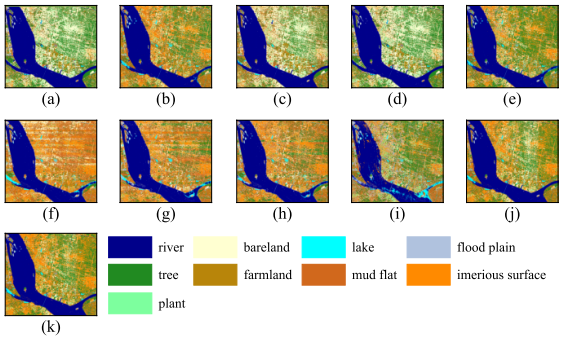

Fig. 6. Thematic maps of Yangtze River Delta (Nantong). (a) LP-CRC. (b) LP-KCRC. (c) LP-ProCRC. (d) Tri 1. (e) Tr 2. (f) GCN. (g) GAT. (h) GraphSAGE. (i) LightGBM. (j) WCD. (k) SSDEL.

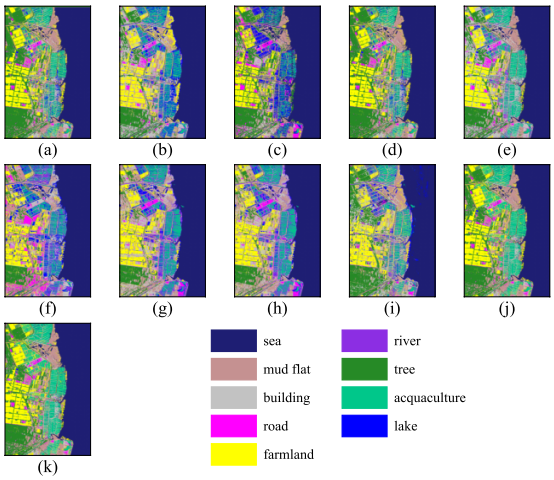

Fig. 7. Thematic maps of Dafeng Natural Reserve. (a) LP-CRC. (b) LP-KCRC. (c) LP-ProCRC. (d) Tri 1. (e) Tr 2. (f) GCN. (g) GAT. (h) GraphSAGE. (i) LightGBM. (j) WCD. (k) SSDEL.

D. Classification Performance and Maps

Tables IV–VI present the accuracy results for three datasets with the same training sample. Thematic maps generated by the proposed algorithms and other comparative models can be observed in Figs. 6–8. The evaluation of classification performance for all algorithms is based on metrics including overall accuracy (OA), average accuracy (AA), and Kappa indexes.

For the Yangtze River Delta dataset result as shown in Table IV and Fig. 6, the OAs of multiple comparative algorithms are reported as 82.07%, 79.68%, 76.36%, 65.52%, 73.81%, 79.09%, 75.85%, 72.98%, 75.89%, and 76.68%, respectively. In contrast, the OAs of the proposed WCD and SSDEL are recorded as 81.72% and 85.18%, respectively. The superior performance of SSDEL is evident. Similarly, SSDEL outperforms in terms of AA and Kappa coefficient, achieving 87.01% and 0.82, respectively.

For the differences in class accuracies, it was observed that in Type 1, the highest accuracy was achieved by tri-training 1 at 99.76%, while in Type 2, accuracies of 98.58% and 98.76% were achieved by LP-CRC and LP-ProCRC, respectively. However, in Type 3, an accuracy of 97.31% was attained by WCD, surpassing other methods. Most notably, in Type 4, significantly higher accuracy was achieved by WCD and SSDEL, reaching 96.00% and 97.77%, respectively, indicating their clear advantages in handling this type of data. Furthermore, for Type 6, much higher accuracy was shown by WCD and SSDEL compared to other methods, with accuracies of 51.52% and 58.80%, respectively, which may reflect their ability to handle challenging data types like Type 6. In Type 7, the top position in accuracy was occupied by SSDEL, with an accuracy of 92.00%, demonstrating superiority in specific contexts. Finally, in Type 9, impressive performances were observed across all methods, with a perfect 100.00% accuracy achieved by LP-ProCRC, tritraining 1, and tritraining 2, while WCD and SSDEL also reached a high accuracy of 99.55%. In summary, the data analysis reveals the advantages of different methods for different types of data. WCD and SSDEL shine in Types 4, 6, and 7, but the effectiveness of each method varies across different types. Therefore, when choosing a method, a customized selection based on the specific data type and task requirements is necessary to achieve the best classification performance.

Table V and Fig. 7 show the results for the second dataset, where the OAs of the five comparison algorithms are reported as 85.93%, 84.35%, 87.68%, 87.28%, 81.51%, 79.09%, 75.85%, 72.98%, 75.89%, and 76.68%, respectively. However, the OAs of the proposed WCD and SSDEL are notably higher, reported as 90.36% and 89.75%, respectively. Moreover, in the provided data analysis, observations were made regarding the accuracy for different classification types (ranging from 1 to 9), and the accuracy advantages of WCD and SSDEL were highlighted in specific instances. Specifically, notable performances of WCD were observed in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
PERFORMANCE OF DIFFERENT METHODS FOR YANGTZE RIVER DELTA

class	*Train*	LP-CRC	LP-KCRC	LP-ProCRC	Tri1	Tri2	GCN	GAT	GraphSAGE	LightGBM	**WCD**	**SSDEL**
1	5	97.86	92.32	90.97	99.76	77.91	91.29	90.58	90.42	81.55	96.82	94.14
2	5	98.58	82.65	98.76	98.05	74.69	48.32	60.00	4.25	76.28	55.08	76.46
3	5	84.74	73.16	91.58	98.95	70.53	92.63	77.37	83.68	73.16	97.31	88.42
4	5	77.09	92.18	94.41	93.85	83.24	94.97	86.59	86.59	78.21	96.00	97.77
5	5	62.99	95.28	99.21	34.65	95.28	99.21	98.43	97.64	91.34	91.87	98.43
6	5	0.86	38.63	0.00	8.58	50.43	50.00	50.43	98.50	67.60	51.52	58.80
7	5	53.88	86.59	82.82	76.24	92.00	88.24	90.35	91.29	92.24	79.81	92.00
8	5	97.10	68.05	59.17	5.21	57.05	76.16	59.46	54.73	53.86	78.00	77.51
9	5	99.55	96.85	100.00	100.00	98.65	94.16	93.71	91.91	99.55	99.55	99.55
OA		82.07	79.68	76.36	65.52	73.81	79.09	75.85	72.98	75.89	81.72	**85.18**
AA		74.74	80.63	80.63	68.37	77.75	81.66	78.55	77.67	79.31	82.88	**87.01**
Kappa		0.78	0.76	0.72	0.60	0.69	0.75	072	0.68	0.72	0.78	**0.82**
Times(s)		2.55	7.30	2.78	1.78	769.78	45.33	25.46	91,01	0.11	38.42	141.00

TABLE V
PERFORMANCE OF DIFFERENT METHODS FOR DAFENG NATURAL RESERVE

class	*Train*	LP-CRC	LP-KCRC	LP-ProCRC	Tri1	Tri2	GCN	GAT	GraphSAGE	LightGBM	**WCD**	**SSDEL**
1	5	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	98.81	100.00	100.00
2	5	95.17	68.73	89.58	95.56	60.81	56.95	54.05	57.72	80.31	97.88	89.96
3	5	22.14	62.86	38.57	27.86	76.43	48.57	60.71	58.57	75.00	40.71	65.71
4	5	98.73	100.00	100.00	100.00	100.00	77.22	100.00	98.73	84.81	100.00	100.00
5	5	86.34	74.45	78.96	87.33	72.69	61.89	77.31	84.03	75.44	88.00	87.44
6	5	16.54	68.05	45.86	37.97	66.17	40.98	70.68	82.71	65.41	33.46	49.25
7	5	97.48	93.14	99.72	99.02	94.96	92.72	94.68	93.70	86.41	99.30	95.80
8	5	90.63	86.82	96.33	79.62	71.88	69.70	88.99	73.10	74.46	94.70	90.22
9	5	18.57	74.29	38.57	94.29	94.29	84.29	64.29	85.71	90.00	94.29	94.29
OA		85.93	84.35	87.68	87.28	81.51	75.19	83.76	83.74	82.7	**90.36**	89.75
AA		69.51	80.92	80.92	80.18	81.91	70.26	78.97	81.59	81.18	83.15	**85.85**
Kappa		0.83	0.81	0.85	0.85	0.78	0.71	0.81	0.81	0.80	**0.88**	**0.88**
Times(s)		3.56	12.07	3.63	2.44	722.04	43.50	17.78	77.20	0.13	26.26	192.63

Classes 2 and 5, achieving high accuracies of 97.88% and 88.00%, respectively, surpassing other methods. Additionally, in Class 7, WCD also demonstrated strong performance, achieving an accuracy of 99.30%, second only to LP-ProCRC (99.72%). On the other hand, SSDEL excelled in Classes 3 and 6, achieving accuracies of 65.71% and 49.25%, respectively, relative to other methods. Overall, the data analysis reveals that WCD and SSDEL exhibit significant accuracy advantages in different class categories although the optimal method varies among different categories, necessitating consideration of specific context and requirements when selecting the method.

Table VI and Fig. 8 are about the results for the last dataset, where the OAs of comparison algorithms are found to be 70.68%, 85.82%, 86.22%, 88.05%, 84.43%, 77.68%, 80.65%, 76.44%, 77.65%, and 86.12%, respectively, while the OAs of WCD and SSDEL are significantly higher at 90.03% and 91.98%, respectively. Furthermore, it is evident that significant advantages are demonstrated by WCD and SSDEL in certain categories. Taking Class 5 as an example, higher accuracy is achieved by WCD and SSDEL, with accuracies of 54.79% and 50.68%, respectively, in comparison to LP-CRC (41.10%) and LP-KCRC (58.90%). This indicates that they are more effective when dealing with data from this specific category. Furthermore, for Class 11, WCD and SSDEL exhibit remarkable advantages with accuracies of 96.50% and 98.90%, significantly surpassing LP-CRC (12.52%) and LP-KCRC (55.62%) in challenging categories. Additionally, for multiple categories, both WCD and SSDEL consistently display high levels of accuracy and achieve higher Kappa values (e.g., 0.89 and 0.91, respectively) in overall performance assessments. Hence, the results on this dataset suggest that WCD and SSDEL can effectively enhance the classification accuracy of specific categories and achieve superior overall performance, particularly when handling challenging categories, where their advantages become more pronounced.

In addition, it is worth highlighting the distinct advantages that WCD and semi-SSDEL exhibit, particularly in scenarios with limited sample sizes. In the context of smaller sample sizes, which can often pose challenges in machine learning tasks, WCD and SSDEL consistently demonstrate robust and superior performance. Moreover, the advantages of WCD and SSDEL are not limited to specific classes, but rather extend across various categories, highlighting their adaptability and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
PERFORMANCE OF DIFFERENT METHODS FOR YELLOW RIVER DELTA

class	*Train*	LP-CRC	LP-KCRC	LP-ProCRC	Tri 1	Tri 2	GCN	GAT	GraphSAGE	LightGBM	WCD	SSDEL
1	5	64.75	80.16	66.06	56.66	81.20	93.47	86.16	60.84	69.97	72.06	74.15
2	5	100.00	100.00	100.00	100.00	84.99	100.00	100.00	98.98	84.35	100.00	99.75
3	5	100.00	100.00	99.01	99.01	69.31	96.04	97.03	100.00	65.35	98.02	99.01
4	5	42.78	88.33	88.33	88.33	68.33	65.56	87.22	72.22	74.44	88.33	88.33
5	5	41.10	58.90	39.73	32.88	68.49	54.79	57.53	30.14	65.75	54.79	50.68
6	5	86.21	75.86	78.16	97.70	66.67	100.00	98.85	98.85	66.67	98.85	97.70
7	5	100.00	95.35	93.02	96.51	90.70	89.53	93.02	94.19	86.05	100.00	100.00
8	5	100.00	99.50	99.50	100.00	97.51	98.51	100.00	100.00	97.01	100.00	100.00
9	5	100.00	93.68	100.00	100.00	89.47	80.00	77.89	95.79	91.05	100.00	99.47
10	5	91.79	95.46	95.03	96.87	88.77	87.90	40.17	47.30	85.53	98.27	97.30
11	5	12.52	55.62	90.24	92.82	59.67	4.24	81.77	41.62	44.57	96.50	98.90
12	5	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.35	100.00	100.00
13	5	78.92	55.56	100.00	100.00	78.63	66.95	61.54	59.54	66.67	91.45	83.19
14	5	11.74	75.22	41.30	38.26	61.30	56.09	63.04	46.09	73.48	47.39	74.35
15	5	86.32	100.00	95.21	100.00	99.32	94.70	97.26	91.45	97.09	99.15	100.00
16	5	57.98	97.08	96.85	95.06	97.75	100.00	100.00	99.55	81.35	98.65	98.88
17	5	21.77	81.45	50.00	57.26	89.52	94.35	40.32	45.16	71.77	61.29	87.90
18	5	29.43	58.04	32.15	56.95	78.47	14.17	79.29	100.00	44.14	54.77	62.67
19	5	96.55	89.66	98.28	100.00	86.21	94.83	98.28	98.28	84.48	100.00	96.55
20	5	0.00	50.00	15.63	0.00	81.25	68.75	78.13	81.25	56.25	21.88	48.44
21	5	6.67	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
OA		70.68	85.82	86.22	88.05	84.43	77.68	80.65	76.44	77.65	90.03	91.98
AA		63.26	83.33	83.33	81.35	82.74	79.04	82.74	79.11	76.44	84.83	88.44
Kappa		0.68	0.85	0.85	0.87	0.83	0.776	0.79	0.75	0.76	0.89	0.91
Times(s)		3.91	18.94	4.83	7.47	1293.63	52.37	148.41	96.47	1.56	147.25	545.03

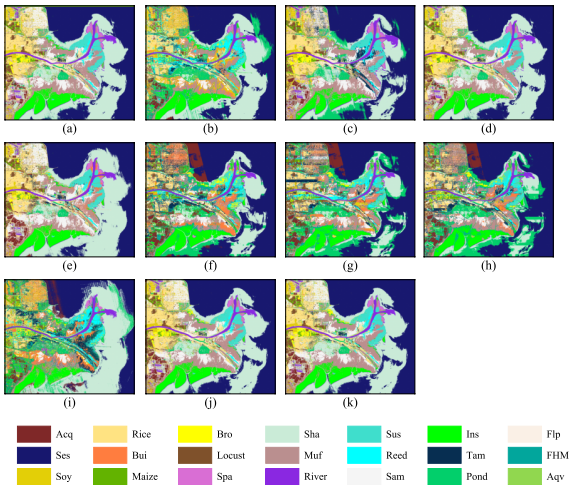

Fig. 8. Thematic maps of Yellow River Delta. (a) LP-CRC. (b) LP-KCRC. (c) LP-ProCRC. (d) Tri 1. (e) Tr 2. (f) GCN. (g) GAT. (h) GraphSAGE. (i) LightGBM. (j) WCD. (k) SSDEL.

reliability in scenarios where data availability may be limited. These results emphasize that WCD and SSDEL are valuable tools in the context of small-sample datasets, where their ability to achieve high accuracy rates and robust performance is especially pronounced. It is worth noting that for the three sets of wetland data, even when compared to several state-of-the-art deep learning and ensemble learning algorithms, the proposed WCD and SSDEL still exhibit a high level of advancement.

In conclusion, superior performance in SSL tasks is demonstrated by the proposed WCD and SSDEL, across the measures of three evaluation metrics, when compared to other algorithms. This validates the efficacy of these novel algorithms in enhancing classification performance.

V. DISCUSSION

A. Analysis of Parameter Sensitivity

Fig. 9(a)–(c) presents the effect of various values for parameters λ_1 and λ_2 on performance in three wetland datasets.

First, for the first wetland dataset, it was observed that the fluctuation in accuracy was negligible for both λ_1 and λ_2. The variation in accuracy was found to be confined between approximately 85.09% and 85.18%, irrespective of the values assigned to λ_1 and λ_2. These findings suggest that the accuracy is minimally influenced by λ_1 and λ_2, thereby highlighting the low sensitivity of the CCO algorithm to parameter selection.

For the second dataset, a similar trend was discerned. The variation in accuracy under different λ_1 and λ_2 values were found to be minimal, with the accuracy rate being consistently maintained between approximately 89.64% and 89.76%. A parallel conclusion was drawn from the Yellow River Delta dataset, where the accuracy varied modestly between approximately 91.80% and 91.98%.

From the aforementioned observations, it can be concluded that the CCO algorithm demonstrates stability across varied wetland datasets. Notably, the algorithm's accuracy remains

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

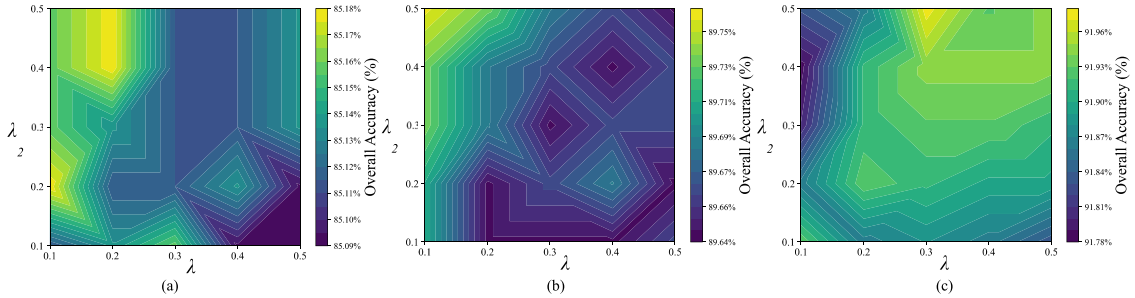

Fig. 9. Effect of parameters λ_1 and λ_2 on OA for CCO. (a) Yangtze River Delta. (b) Jiangsu Dafeng Natural Reserve. (c) Yellow River Delta.

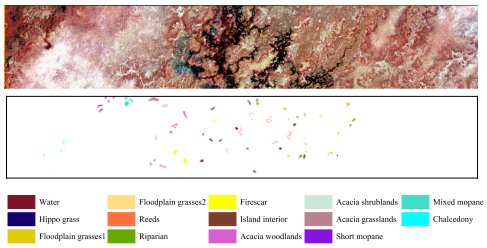

Fig. 10. Botswana dataset.

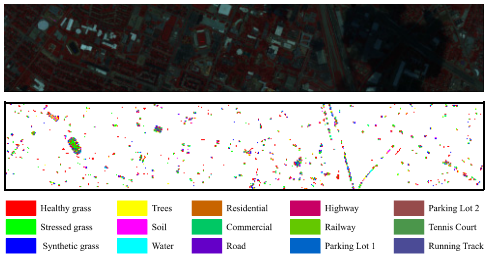

Fig. 11. Houston dataset.

relatively consistent, regardless of the values assigned to λ_1 and λ_2, corroborating the algorithm's stability.

In conclusion, a comprehensive analysis of three wetland datasets led to the finding that the CCO algorithm possesses qualities of stability and practicality. Regardless of the dataset and parameter values employed, the CCO algorithm can yield consistently reliable and comparable accuracy results. This mitigates the necessity for extensive parameter tuning, thereby enhancing the algorithm's reliability and practicality.

B. Generalizability Analysis

To more comprehensively evaluate the generalizability of the proposed WCD and SSDEL algorithms, two additional datasets, the Botswana wetland dataset and the Houston HIS dataset, are introduced. These two datasets, collected by different sensors, present notable variations and distinct characteristics compared to the previous three datasets. By conducting performance evaluations on these distinct and diverse datasets, a better understanding and demonstration of the ability and robustness of the new algorithms in dealing with varied and complex data can be achieved.

The Botswana wetland dataset as shown in Fig. 10 was sourced from the Okavango Delta in Botswana, gathered by the NASA EO-1 satellite in May 2001. For the Botswana wetland dataset, comparisons were also made between the ten comparative algorithms, and the proposed WCD and SSDEL. On this dataset (Table VII), the OA values of the nine algorithms were 78.67%, 80.76%, 82.65%, 79.96%, 71.54%, 73.58%, 83.39%, 80.02%, and 65.85%, respectively, while the OAs of the proposed WCD and SSDEL were 85.38% and 88.15%, respectively, significantly outperforming other algorithms. In terms of AA and Kappa coefficients, WCD and SSDEL also performed remarkably, with AAs reaching 87.30% and 89.75%, and Kappa coefficients at 0.84 and 0.87, respectively.

The Houston data were acquired by the ITRES CASI-1500 sensor and provided by the 2013 IEEE GRSS Data Fusion Contest. The data have a size of 349 × 1905 pixels and consist of 144 spectral bands ranging from 364 to 1046 nm. Land cover is annotated with 15 different categories, as shown in Fig. 11. For this dataset, comparisons were also made between the ten comparative algorithms, and the proposed WCD and SSDEL. On this dataset (Table VIII), the OA values of the ten algorithms were 58.06%, 75.00%, 64.43%, 69.33%, 75.50%, 66.90%, 75.70%, 75.80%, and 74.47%, respectively, while the OAs of the proposed WCD and SSDEL were 77.04% and 77.04%, respectively, significantly outperforming other algorithms.

These results further validate the broad adaptability and performance of the new algorithms on wetland data collected from different sensors, confirming their high generalizability.

C. Effectiveness of the Proposed Semi-Supervised Algorithm

1) Extensibility of Augmented Samples: To validate the efficacy of the proposed sample augmentation algorithm, a comprehensive set of comparative experiments was undertaken. The intent was to contrast the classification accuracy derived from original training samples against those resulting from augmented samples across a variety of classifiers. Augmented samples 1, Augmented samples 2, and Augmented samples 3 correspond to the augmented samples generated from different classifier combinations proposed in our semi-supervised algorithm.

From Fig. 12(a)–(c), for the Yangtze River Delta dataset, the OA range of the original training samples was found to be between 69.50% and 81.03%. However, upon the integration of any of the augmented samples, a notable increase in OA was observed, reaching up to 84.15%. Similar trends were witnessed with the Jiangsu Dafeng Natural Reserve and Yellow

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
CLASSIFICATION PERFORMANCE FOR BOTSWANA DATASET

class	Train	LP-CRC	LP-KCRC	LP-ProCRC	Tri1	Tri2	GCN	GAT	GraphSAGE	LightGBM	**WCD**	**SSDEL**
1	5	61.92	99.62	100.00	93.08	97.69	100.00	100.00	100.00	100.00	98.46	91.92
2	5	70.65	98.91	98.91	100.00	95.65	100.00	98.91	100.00	83.70	100.00	100.00
3	5	93.39	88.02	100.00	98.76	84.71	86.78	86.78	90.50	77.69	96.28	92.98
4	5	97.07	91.71	85.37	99.02	75.61	70.24	82.93	73.17	42.93	98.54	98.05
5	5	86.87	76.45	61.00	1.16	45.95	19.31	68.34	51.35	49.42	53.28	83.01
6	5	31.27	49.81	74.13	67.57	37.07	66.80	62.93	63.32	49.03	74.52	58.69
7	5	93.98	89.96	98.80	98.80	89.96	98.39	95.58	96.79	65.46	99.20	98.80
8	5	98.97	97.42	88.66	97.42	82.99	88.14	80.41	88.14	46.39	97.42	99.48
9	5	70.72	60.53	77.96	84.21	65.79	73.36	79.61	76.97	64.14	77.30	80.92
10	5	59.66	68.07	43.70	32.77	69.33	42.86	79.41	63.45	60.08	60.50	69.75
11	5	98.98	85.08	93.56	98.64	82.03	91.86	90.17	95.25	81.02	98.64	97.97
12	5	97.66	97.08	100.00	94.74	63.16	88.89	88.30	87.13	73.68	97.08	98.83
13	5	63.71	67.57	62.93	88.03	47.88	4.63	76.83	62.55	54.44	72.20	87.26
14	5	98.84	98.84	100.00	98.84	100.00	98.84	97.67	97.67	98.84	98.84	98.84
OA		78.67	80.76	82.65	79.96	71.54	73.58	83.39	80.02	65.85	85.38	**88.15**
AA		80.26	83.50	83.50	82.36	74.13	70.35	84.85	81.88	67.63	87.30	**89.75**
Kappa		0.77	0.79	0.81	0.78	0.69	0.68	0.82	0.78	0.63	0.84	**0.87**

TABLE VIII
CLASSIFICATION PERFORMANCE FOR HOUSTON DATASET

class	Train	LP-CRC	LP-KCRC	LP-ProCRC	Tri1	Tri2	GCN	GAT	GraphSAGE	LightGBM	**WCD**	**SSDEL**
1	20	80.43	95.62	75.72	88.11	89.93	95.95	96.61	96.37	85.96	95.54	95.21
2	20	86.41	91.27	81.38	94.07	98.85	95.06	98.43	98.52	96.21	98.68	99.01
3	20	99.70	96.04	100.00	99.85	90.11	98.48	97.87	98.02	90.72	96.50	96.04
4	20	93.20	91.71	98.34	96.43	91.63	93.03	93.37	93.20	88.64	92.79	92.79
5	20	89.62	90.53	99.92	99.92	84.22	78.74	80.07	87.13	78.32	96.18	89.62
6	20	85.26	96.84	96.49	89.27	87.02	90.88	94.39	94.39	90.53	97.54	97.19
7	20	42.51	73.21	79.80	64.01	75.98	68.16	75.24	74.67	73.86	73.21	75.41
8	20	48.17	48.75	50.58	61.83	51.99	5.15	57.14	61.05	48.34	47.43	49.09
9	20	94.64	78.80	73.51	85.66	63.86	33.09	57.34	75.08	69.47	72.61	68.32
10	20	0.00	49.16	5.72	33.67	64.39	28.03	42.42	40.74	63.97	64.14	69.02
11	20	1.67	64.63	14.38	0.00	71.15	63.46	61.20	56.69	70.07	61.62	65.97
12	20	0.00	44.30	24.96	37.86	55.78	73.70	78.48	62.90	57.87	51.09	53.60
13	20	26.74	36.05	42.09	26.45	26.74	26.98	23.72	8.37	32.56	32.33	36.28
14	20	66.75	95.10	67.53	96.14	90.98	94.33	94.85	96.13	88.14	96.39	96.65
15	20	99.20	97.91	99.04	99.04	91.64	98.71	98.71	98.39	92.77	97.75	97.91
OA		58.06	75.00	64.43	69.33	75.50	66.90	75.70	75.80	74.47	77.04	**77.04**
AA		60.95	76.66	76.66	71.49	75.62	69.58	76.66	76.11	75.16	78.25	**78.81**
Kappa		0.55	0.73	0.62	0.67	0.73	0.64	0.74	0.74	0.72	0.75	**0.76**

River Delta datasets, which further reinforces the positive influence of the augmentation process implemented in this article.

These findings convincingly demonstrated the advantages of the proposed SSL approach in enhancing classification accuracy. The consistent improvement across different datasets and algorithms signify the robustness and versatility of the proposed algorithm.

2) Effectiveness of Augmented Samples: From the results in Table IX, it can be seen that the SSL algorithms proposed for data augmentation, namely, WCD 1, WCD 2, and WCD 3, consistently outperform traditional approaches such as LP, Tr1, and Tr2 across multiple datasets.

For the first dataset, the classification accuracy achieved by LP, Tr1, and Tr2 after data augmentation are 0.79, 0.66, and 0.74, respectively, whereas the proposed algorithms WCD 1, WCD 2, and WCD 3 achieve accuracies of 0.95, 0.94, and 0.91, respectively. This clearly demonstrates the superior performance of the proposed WCD series algorithms on this dataset.

TABLE IX
ACCURACY OF THE PROPOSED WCD VERSUS OTHER SSL METHODS

Sample Augmentation Algorithms	LP	Tri1	Tri2	WCD 1	WCD 2	WCD 3
Yangtze River Delta	0.79	0.66	0.74	**0.95**	0.94	0.91
Jiangsu Dafeng Natural Reserve	0.84	0.87	0.82	**0.89**	0.80	0.88
Yellow River Delta	0.88	0.88	0.84	**0.96**	0.88	**0.96**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

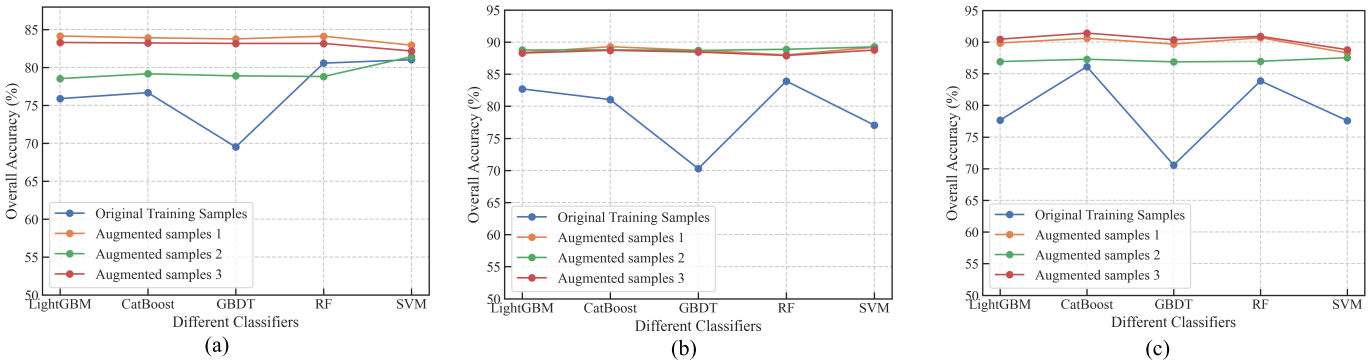

Fig. 12. Extensibility of augmented samples with different classifiers. (a) Yangtze River Delta. (b) Jiangsu Dafeng Natural Reserve. (c) Yellow River Delta.

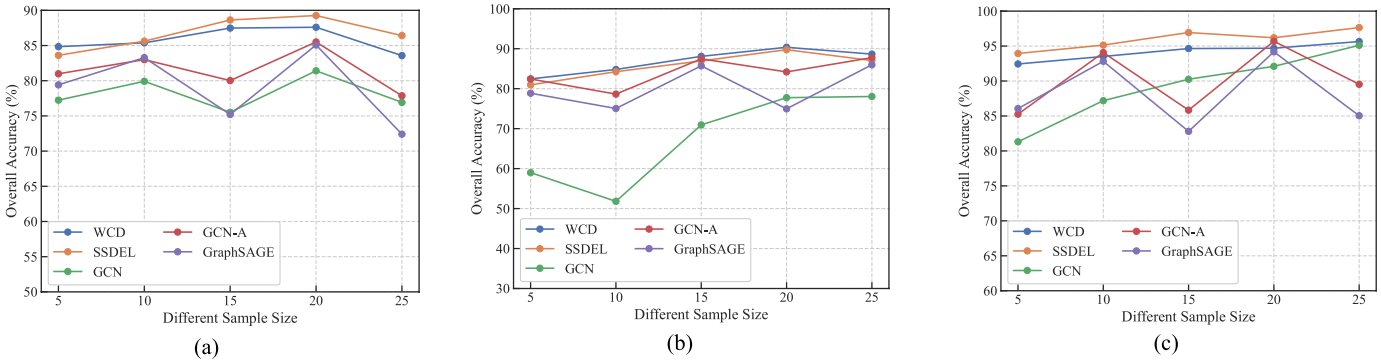

Fig. 13. Performance of the proposed methods versus state-of-the-art deep learning algorithms based on different sample sizes. (a) Yangtze River Delta. (b) Jiangsu Dafeng Natural Reserve. (c) Yellow River Delta.

TABLE X
ABLATION ANALYSIS

Algorithm Description	Based Ensemble Learning (with no pseudo label)	√	√	√	√
	Feature Weighting		√	√	√
	Cosine Similarity			√	√
	Consistent Coefficient Optimization				√
Datasets	Yangtze River Delta (OA%)	79.02	80.15	83.26	**85.18**
	Jiangsu Dafeng Natural Reserve (OA%)	87.95	89.43	**90.36**	89.75
	Yellow River Delta (OA%)	86.15	87.98	90.03	**91.98**

On the Jiangsu Dafeng Natural Reserve dataset, the classification accuracies of LP, Tr1, and Tr2 after augmentation are 0.84, 0.87, and 0.82, respectively. Although WCD 2 demonstrates a slightly lower accuracy of 0.80, WCD 1 and WCD 3 outperform the traditional methods with accuracies of 0.89 and 0.88, respectively.

Finally, on the Yellow River Delta dataset, traditional algorithms LP, Tr1, and Tr2 yield accuracies of 0.88, 0.88, and 0.84 after data augmentation, whereas the proposed WCD 1, WCD 2, and WCD 3 algorithms deliver significantly higher accuracies of 0.96, 0.88, and 0.96, respectively.

Overall, the presented data support the efficacy of the proposed WCD series algorithms in SSL tasks for data augmentation. The novel algorithms exhibit robust generalization capabilities across diverse datasets and consistently demonstrate superior or comparable performance to traditional methods. In most scenarios, the WCD series algorithms noticeably surpass the LP, Tr1, and Tr2 methods, emphasizing their advantage in enhancing data augmentation accuracy.

D. Performance of Proposed Algorithms Versus Other State-of-the-Art Algorithms Based on Different Sample Sizes

Fig. 13 shows the classification accuracy results of the proposed algorithm compared to other comparative algorithms under different sample sizes.

In this experiment, several state-of-the-art deep learning algorithms were compared with the proposed WCD and SSDEL algorithms under different sample conditions. The sample size is set to 5, 10, 15, 20, and 25, respectively.

Fig. 13(a) shows the results of Yangtze River Delta data. The proposed algorithm can achieve the highest accuracy regardless of the change in sample size compared to the other three deep learning algorithms. Furthermore, the classification accuracy of the proposed algorithm varies more smoothly with the number of samples and has better stability compared to other algorithms. For the two sets of data, Jiangsu Dafeng Natural Reserve and Yellow River Delta, similar conclusions can be drawn from Fig. 13(b) and (c). In summary, even in different sample sizes, the proposed algorithm still has the highest accuracy and is more stable.

E. Ablation Analysis

From the information in Table X, a series of ablation analyses was performed to evaluate the superiority of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

proposed method. First, on the basis of the basic ensemble learning model (without pseudo labels), the gradual introduction of different improvement methods significantly affected the model's performance. On the Yangtze River Delta dataset, the introduction of Feature Weighting resulted in a 1.13% improvement in accuracy, while on the Jiangsu Dafeng Natural Reserve dataset and Yellow River Delta dataset, it increased accuracy by 1.48% and 1.83%, respectively. This indicates that the introduction of feature weighting effectively enhances the model's utilization of features, thereby improving its performance.

Second, the model's performance saw another significant boost with the further introduction of cosine similarity. On the Yangtze River Delta dataset, an additional 3.11% improvement was achieved, while on the Jiangsu Dafeng Natural Reserve dataset, it increased by 0.93%, and on the Yellow River Delta dataset, it improved by 2.05%. This demonstrates the effectiveness of cosine similarity in enhancing the model's performance, especially on the Yangtze River Delta and Yellow River Delta datasets.

Finally, the introduction of consistent coefficient optimization continued to improve the model's performance. On the Yangtze River Delta dataset, it increased by 1.92%, on the Jiangsu Dafeng Natural Reserve dataset by 0.39%, and on the Yellow River Delta dataset by 1.95%. This suggests that consistent coefficient optimization is helpful in further enhancing the model's performance, particularly on the Yangtze River Delta and Yellow River Delta datasets.

In summary, the experimental results clearly demonstrate the effectiveness of gradually introducing improvement methods such as feature weighting, cosine similarity, and consistent coefficient optimization in enhancing model performance.

VI. Conclusion

In conclusion, a novel semi-supervised ensemble learning framework called SSDEL combining pseudo-label generation, sample augmentation, and consistent learning is introduced for wetland classification in this article. The proposed techniques overcome the limitations of labeled samples and challenges posed by ensemble learning. However, it is important to note that this method has some limitations. First, pseudo-label generation may introduce some erroneous labels, especially in cases of poor data quality or uneven sample distribution. Second, the effectiveness of the sample augmentation method may be influenced by the characteristics of the dataset and the diversity of wetland types. In future research, we plan to incorporate transfer learning theory and conduct experiments on larger scale datasets to address the aforementioned limitations. Despite these limitations, the experimental results still demonstrate that SSDEL outperforms state-of-the-art methods, showcasing its effectiveness in wetland classification. The extensibility of SSDEL is demonstrated by achieving better results on multiple machine learning models with new samples. Moreover, its generalizability is evidenced by the outstanding performance on the Botswana dataset. Overall, the proposed SSDEL framework demonstrates the potential for enhancing wetland classification accuracy and efficiency, suggesting opportunities for future research and practical applications in this domain.

References

[1] B. Du et al., "Mapping wetland plant communities using unmanned aerial vehicle hyperspectral imagery by comparing object/pixel-based classifications combining multiple machine-learning algorithms," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8249–8258, 2021.

[2] T. Schmid, M. Koch, and J. Gumuzzio, "Multisensor approach to determine changes of wetland characteristics in semiarid environments (central Spain)," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 11, pp. 2516–2525, Nov. 2005.

[3] Q. He, X. Sun, Z. Yan, B. Li, and K. Fu, "Multi-object tracking in satellite videos with graph-based multitask modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619513.

[4] S. Bhatnagar, L. Gill, S. Regan, S. Waldren, and B. Ghosh, "A nested drone-satellite approach to monitoring the ecological conditions of wetlands," *ISPRS J. Photogramm. Remote Sens.*, vol. 174, pp. 151–165, Apr. 2021.

[5] A. Hoblitzell, M. Babbar-Sebens, and S. Mukhopadhyay, "Uncertainty-based deep learning networks for limited data wetland user models," in *Proc. IEEE Int. Conf. Artif. Intell. Virtual Reality (AIVR)*, Dec. 2018, pp. 19–26.

[6] Y. Gao et al., "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512615, doi: 10.1109/TGRS.2021.3097093.

[7] K. Liu et al., "Mapping coastal wetlands using transformer in transformer deep network on China ZY1-02D hyperspectral satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3891–3903, 2022.

[8] J. Li, H. Zhang, and L. Zhang, "A nonlinear multiple feature learning classifier for hyperspectral images with limited training samples," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2728–2738, Jun. 2015.

[9] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.

[10] Y. Su et al., "ACGT-net: Adaptive cuckoo refinement-based graph transfer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521314.

[11] Y. Su, M. Jiang, L. Gao, X. You, X. Sun, and P. Li, "Graph-cut-based node embedding for dimensionality reduction and classification of hyperspectral remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Kuala Lumpur, Malaysia, Jul. 2022, pp. 1720–1723.

[12] T. Xiao, Y. Liu, Y. Huang, M. Li, and G. Yang, "Enhancing multiscale representations with transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605116.

[13] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[14] B. Tu, X. Liao, Q. Li, Y. Peng, and A. Plaza, "Local semantic feature aggregation-based transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536115.

[15] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[16] Y. Gao, M. Zhang, J. Wang, and W. Li, "Cross-scale mixing attention for multisource remote sensing data fusion and classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507815.

[17] H. Yu, Z. Ling, K. Zheng, J. Li, S. Liang and L. Gao, "Unsupervised dynamic convolutional neural network model for hyperspectral and multispectral image fusion," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 6270–6273

[18] X. Zhang, Y. Su, L. Gao, L. Bruzzone, X. Gu, and Q. Tian, "A lightweight transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517617.

[19] P. Liu, B. Fu, S. X. Yang, L. Deng, X. Zhong, and H. Zheng, "Optimizing survival analysis of XGBoost for ties to predict disease progression of breast cancer," *IEEE Trans. Biomed. Eng.*, vol. 68, no. 1, pp. 148–160, Jan. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

[20] Y. Wang, J. Sha, L. Gao, Y. Zhang, X. Rong, and C. Zhang, "A semi-supervised domain alignment transformer for hyperspectral images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525411.

[21] Y. Ding, X. Zhao, Z. Zhang, W. Cai, N. Yang, and Y. Zhan, "Semi-supervised locality preserving dense graph neural network with ARMA filters and context-aware learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511812, doi: 10.1109/TGRS. 2021.3100578.

[22] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[23] H. Wu and S. Prasad, "Semi-supervised deep learning using pseudo labels for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1259–1270, Mar. 2018.

[24] H. Su, F. Shao, Y. Gao, H. Zhang, W. Sun, and Q. Du, "Probabilistic collaborative representation based ensemble learning for classification of wetland hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509517.

[25] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

[26] A. Qin et al., "Distance constraint-based generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511416, doi: 10.1109/TGRS.2023.3274778.

[27] M. S. Kotzagiannidis and C.-B. Schönlieb, "Semi-supervised superpixel-based multi-feature graph learning for hyperspectral image data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4703612.

[28] Y. Zhang, G. Cao, B. Wang, X. Li, P. Y. O. Amoako, and A. Shafique, "Dual sparse representation graph-based copropagation for semisupervised hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500517.

[29] X. Zhang, Q. Song, R. Liu, W. Wang, and L. Jiao, "Modified Co-training with spectral and spatial views for semisupervised hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2044–2055, Jun. 2014.

[30] W. Wang, S. Wan, P. Xiao, and X. Zhang, "A novel multi-training method for time-series urban green cover recognition from multitemporal remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9531–9544, 2022.

[31] J. Li, S. Wu, C. Liu, Z. Yu, and H.-S. Wong, "Semi-supervised deep coupled ensemble learning with classification landmark exploration," *IEEE Trans. Image Process.*, vol. 29, pp. 538–550, 2020.

[32] T.-H. Lin and C.-H. Lin, "Hyperspectral change detection using semi-supervised graph neural network and convex deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515818.

[33] H. Su, Y. Yu, Q. Du, and P. Du, "Ensemble learning for hyperspectral image classification using tangent collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3778–3790, Jun. 2020.

[34] D. Cao, M. Zhang, W. Li, and Q. Ran, "Hyperspectral and infrared image collaborative classification based on morphology feature extraction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4405–4416, 2021.

[35] Z. Wu et al., "Hyperspectral anomaly detection with relaxed collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533417.

[36] W. Li, Q. Du, and M. Xiong, "Kernel collaborative representation with Tikhonov regularization for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 1, pp. 48–52, Jan. 2015.

[37] S. Bahrampour, N. M. Nasrabadi, A. Ray, and K. W. Jenkins, "Kernel task-driven dictionary learning for hyperspectral image classification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2015, pp. 1324–1328.

[38] J. Liu, Z. Wu, J. Li, A. Plaza, and Y. Yuan, "Probabilistic-kernel collaborative representation for spatial–spectral hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2371–2384, Apr. 2016.

[39] X. Feng, Z. Xiao, B. Zhong, J. Qiu, and Y. Dong, "Dynamic ensemble classification for credit scoring using soft probability," *Appl. Soft Comput.*, vol. 65, pp. 139–151, Apr. 2018.

[40] L. Du, R. Gao, P. N. Suganthan, and D. Z. W. Wang, "Bayesian optimization based dynamic ensemble for time series forecasting," *Inf. Sci.*, vol. 591, pp. 155–175, Apr. 2022.

[41] R. M. O. Cruz, H. H. Zakane, R. Sabourin, and G. D. C. Cavalcanti, "Dynamic ensemble selection VS K-NN: Why and when dynamic selection obtains higher classification performance?" in *Proc. 7th Int. Conf. Image Process. Theory, Tools Appl. (IPTA)*, Nov. 2017, pp. 1–6.

[42] H. Lu, H. Su, J. Hu, and Q. Du, "Dynamic ensemble learning with multi-view kernel collaborative subspace clustering for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2681–2695, 2022.

[43] R. M. O. Cruz, R. Sabourin, and G. D. C. Cavalcanti, "META-DES.H: A dynamic ensemble selection technique using meta-learning and a dynamic weighting approach," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Killarney, Ireland, 2015, pp. 1–8.

[44] R. M. O. Cruz, R. Sabourin, and G. D. C. Cavalcanti, "Dynamic classifier selection: Recent advances and perspectives," *Inf. Fusion*, vol. 41, pp. 195–216, May 2018.

[45] S. Zhong et al., "Class feature weighted hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4728–4745, Dec. 2019.

[46] C. Liu, J. Zhou, J. Liang, Y. Qian, H. Li, and Y. Gao, "Exploring structural consistency in graph regularized joint spectral–spatial sparse coding for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 3, pp. 1151–1164, Mar. 2017.

[47] A. Samat, E. Li, P. Du, S. Liu, and J. Xia, "GPU-accelerated CatBoost-forest for hyperspectral image classification via parallelized mRMR ensemble subspace feature selection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3200–3214, 2021.

[48] X. Sun, L. Guo, W. Zhang, Z. Wang, and Q. Yu, "Small aerial target detection for airborne infrared detection systems using LightGBM and trajectory constraints," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 9959–9973, 2021.

[49] Z. Zhang and C. Jung, "GBDT-MO: Gradient-boosted decision trees for multiple outputs," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 7, pp. 3156–3167, Jul. 2021.

[50] X. He, Y. Chen, and P. Ghamisi, "Dual graph convolutional network for hyperspectral image classification with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502418, doi: 10.1109/TGRS.2021.3061088.

[51] A. Sha, B. Wang, X. Wu, and L. Zhang, "Semisupervised classification for hyperspectral images using graph attention networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 157–161, Jan. 2021.

[52] Y. Cui, C. Shao, L. Luo, L. Wang, S. Gao, and L. Chen, "Center weighted convolution and GraphSAGE cooperative network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508216, doi: 10.1109/TGRS.2023.3264653.

Hongliang Lu received the B.E. degree from Chuzhou University, Chuzhou, China, in 2015, and the M.E. degree from the School of Geomatics, Anhui University of Science and Technology, Huainan, China, in 2020. He is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the School of Earth Sciences and Engineering, Hohai University, Nanjing, China.

His research interests include ensemble learning, few shot learning, and collaborative representation in hyperspectral remote sensing imagery classification.

Hongjun Su (Senior Member, IEEE) received the Ph.D. degree in cartography and geography information system from the Key Laboratory of Virtual Geographic Environment (Ministry of Education), Nanjing Normal University, Nanjing, China, in 2011.

He is currently a Full Professor with the School of Earth Sciences and Engineering, Hohai University, Nanjing. His main research interests include hyperspectral remote sensing dimensionality reduction, classification, and spectral unmixing.

Dr. Su is an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.

Pan Zheng received the B.E. degree in geodesy and geomatics engineering and the M.E. degree in photogrammetry and remote sensing from the School of Earth Sciences and Engineering, Hohai University, Nanjing, China, in 2018 and 2021, respectively, where she is currently pursuing the Ph.D. degree in surveying and mapping.

Her research interests include collaborative representation and machine learning in hyperspectral unmixing.

Yihan Gao received the B.E. degree from Hohai University, Nanjing, China, in 2020, where she is currently pursuing the M.E. degree in surveying and mapping engineering with the School of Earth Sciences and Engineering.

Her research interests include hyperspectral remote sensing and soil moisture retrievals from remote sensing.

Hengyi Zheng is currently pursuing the B.E. degree in geomatics engineering with the School of Earth Sciences and Engineering, Hohai University, Nanjing, China.

Her research interests include collaborative representation and hyperspectral remote sensing imagery classification.

Zhaohui Xue (Member, IEEE) received the B.S. degree in geomatics engineering from Shandong Agricultural University, Tai'an, China, in 2009, the M.E. degree in remote sensing from China University of Mining and Technology, Beijing, China, in 2012, and the Ph.D. degree in cartography and geographic information system from Nanjing University, Nanjing, China, in 2015.

He is currently a Professor with the School of Earth Sciences and Engineering, Hohai University, Nanjing. His research interests include hyperspectral image classification, time-series image analysis, pattern recognition, and machine learning.

Weiwei Sun (Senior Member, IEEE) received the B.S. degree in surveying and mapping and the Ph.D. degree in cartography and geographic information engineering from Tongji University, Shanghai, China, in 2007 and 2013, respectively.

He is currently a Full Professor with the Department of Geography and Spatial Information Techniques, Ningbo University, Ningbo, Zhejiang, China. His research interests include hyperspectral image processing with machine learning.

Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore, MD, USA, in 2000.

She is currently the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of the SPIE-International Society for Optics and Photonics. From 2016 to 2020, she was the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She is currently a member of IEEE Periodicals Review and Advisory Committee.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:34:16 UTC from IEEE Xplore. Restrictions apply.